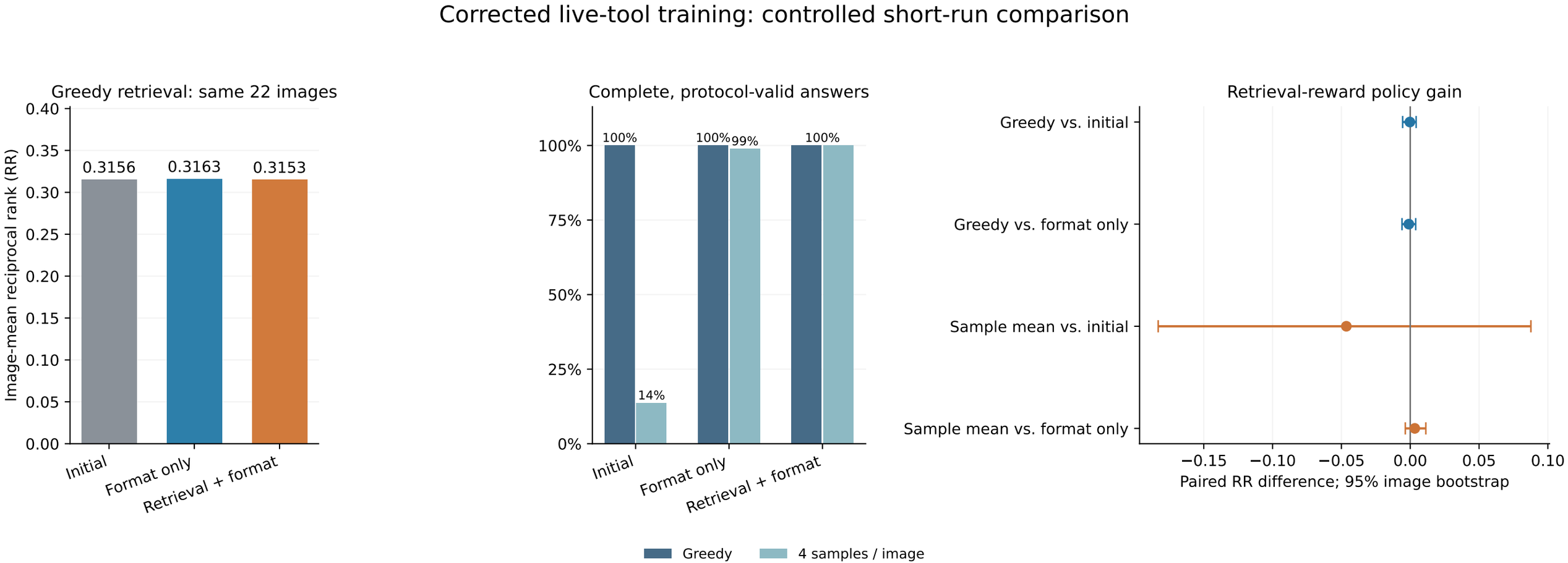
Corrected live-tool training: controlled short-run comparison
Greedy retrieval: same 22 images
Image-mean reciprocal rank (RR)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.3156
0.3163
0.3153
Initial
Format only
Retrieval + format
Complete, protocol-valid answers
100%
75%
50%
25%
0%
100%
14%
100%
99%
100%
Initial
Format only
Retrieval + format
Retrieval-reward policy gain
Greedy vs. initial
Greedy vs. format only
Sample mean vs. initial
Sample mean vs. format only
−0.15
−0.10
−0.05
0.00
0.05
0.10
Paired RR difference; 95% image bootstrap
Greedy
4 samples / image
20 planned prompt groups per arm; actual optimizer updates are recorded separately. Frozen final retrieval evaluator.